Fiscal Year:	FY 2016	**Task Last Updated:**	FY 10/12/2017
PI Name:	Hollberg, Leo W Ph.D.		
Project Title:	Advancing Time Transfer and Optical Atomic Clocks for Space		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Stanford	**State:**	CA
Zip Code:	94305-4008	**Congressional District:**	15
Comments:	NOTE: Previously at National Institute of Standards & Technology (Ed., Fall 2011)		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	05/30/2012	**End Date:**	08/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2017 per JPL info and PI (Ed., 9/22/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 1458931		
Performance Goal No.:			
Performance Goal Text:			
	Our research project focuses on one of the most critical and significant scientific and technical challenges to using advanced cold-atom atomic clocks in space. The problems that we address are the serious performance limitations to the existing Time-Transfer links between ground and space. This research will advance both the basic science and the technology of Time-Transfer methods that could both support the current European Space Agency-Atomic Clock Ensemble in Space (ESA-ACES) mission plan and subsequently enhance performance in order to meet the demands of future optical atomic clocks (e.g., ESAs-SOC) for space. The improved performance is enabled by advanced laser methods for absolute Time and Frequency transfer that are based on stabilized femtosecond mode-locked lasers. We have developed a novel integrated system concept that will optimize the Time-Transfer performance in real-world systems by combining an ultrahigh performance Laser Time-Transfer (Laser-TT) link with an advanced microwave link		

Task Description:	and GPS signals. Analysis of the fundamental capabilities and limitations will guide the final system design, and a prototype Laser-TT link will be built in the laboratory and tested to critically evaluate the system performance. The demonstrations and testing of the Time-Transfer link will include two-way laser transmission via both free-space and optical fibers, and two-way microwave transmission via coaxial cable, which will give over-constrained redundant signals allowing us to identify systematic errors and uncertainties. Our Laser-TT link will improve the absolute timing precision by a factor of 100 times compared to the most advanced microwave links. Nonetheless, the highest performance, practical, and reliable systems operating in real-world conditions (with weather, turbulence, etc.) will require both microwave and optical channels. Critical analysis and testing of the most promising concepts will allow us to optimally integrate an advanced microwave link (as per ACES) with our higher performance Laser-TT link.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced precision for "Time" (epoch), time interval, and frequency transfer to space could be enabling for several science and technology missions and applications, including: International Space Station (ISS), ACES, tests of relativity, geodesy, gravity. Optical timing can improve the performance and precision of Global Satellite Navigation Systems (GNSS) such as GPS. It could also provide much improved precision and accuracy of precise orbit determination and through better timing and orbit could enhance capabilities and accuracy in geodesy. Potentially leverage capabilities with existing International Laser Ranging Systems to mutual benefit. Definitely strong synergism with satellite laser communication systems now under development.
Task Progress:	Research Activities and Progress during the Fourth Year of Funding 1) Laser Communication links for Time/frequency transfer We have continued to develop an experimental setup to measure the phase noise and Time-transfer stability of a model space lasercom system. In keeping with our goal to utilize technologies compatible with NASA's current and planned space lasercom missions our system emulates a 1 GHz Return-to-Zero Differential Phase Shift Keying (RZ-DPSK) communications system operating at 1550 nm with existing telecom components (CITE). Our goal is to measure the Timing stability of clock recovery techniques used to recover the encoded data in the DPSK system and develop and test new techniques to provide the stability required for our 1ps Timing goal. We are testing our setup by measuring the round trip phase noise on a 4km fiber loop on campus. This loop was previously measured using the RF repetition rate and optical pulse interference from a femtosecond optical frequency comb as reported in our 2014 report. Our new measurements are based on the RF mixer method comparing the phase of the outgoing and incoming signal envelopes. In order to explore the Time-transfer performance at the low signal powers expected at a satellite receiver we use a variable attenuator to reduce the optical power incident on the receiver to 10 nW, followed by an Erbium-Doped Fiber Amplifier (EDFA) to boost the received signal for detection. This setup is similar to a satellite optical receiver front end and allows us to measure the effect of Amplified Spontaneous Emission (ASE) noise on the Time-transfer link. The EDFA operates at 45 dB of gain. As of August 2016 we have constructed a Mach-Zehnder delay line interferometer operating at a symbol rate of 860 MHz. A future version will be measured for the target 1 GHz symbol rate. The delay line is stabilized with a custom PZT fiber stretcher to maintain correct optical carrier interference at the constructive and destructive output ports. We are currently sourcing a suitable digitizer to recover the demodulated bit stream and complete our RZ-DPSK receiver. With the completed system we will be able to test different methods of transmitting Time epoch using the DPSK bit stream, as well as different timing recovery techniques. 2) STR concept proposal paper Our high performance Space Time Reference (STR) concept paper was published online in Classical and Quantum Gravity in May 2016 [see Bibliography section]. As detailed last year this STR system is based on a satellite in a MEO orbit that could deliver high accuracy Time/frequency and accurate range information to essentially anywhere on the Earth. The ground stations would connect high accuracy clocks on the ground to the STR using two-way lasercom methods. Our goal continues to be 1 ps timing and 1 mm range information between the ground stations and the STR satellite. 3) Limits of atmospheric non-reciprocity on optical Time-frequency transfer to space Prof. Aniceto Belmonte from PTC Barcelona has been collaborating with us on this project as an expert on free space optics and the effects of atmospheric turbulence. Over this last year he has performed an analysis on the limits of partial reciprocity on Time-transfer stability using optical links.
Bibliography Type:	Description: (Last Updated: 09/03/2025)
Articles in Peer-reviewed Journals	Berceau P, Taylor M, Kahn J, Hollberg L. "Space-time reference with an optical link." Classical and Quantum Gravity. 2016 Jul 7;33(13):135007. Published online May 31, 2016. http://dx.doi.org/10.1088/0264-9381/33/13/135007 , Jul-2016